

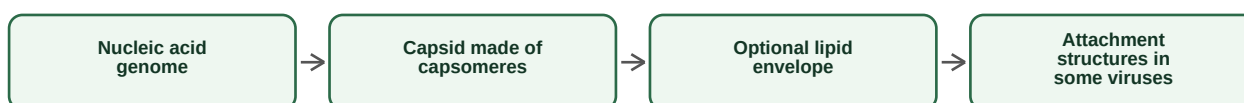
UNIT II — VIRUS AND PROKARYOTES

2.1 Viruses: history and general account

Definition — Virus: An acellular infectious entity containing a nucleic-acid genome enclosed in a protein coat, and in some cases an additional envelope. Viruses replicate only inside suitable host cells.

Feature	Virus	Typical cellular organism
Cellular organization	Acellular	Cellular
Independent metabolism	Absent	Present in some form
Replication	Uses host-cell machinery	Cell division or cellular reproduction
Genetic material	DNA or RNA genome	Usually both DNA and RNA in cellular systems

Major structural components of a virus



- **Genome:** carries genetic information.
- **Capsid:** protein coat that protects the genome and may assist in host interaction.
- **Envelope:** host-derived lipid membrane present in some viruses.
- **Host specificity:** attachment depends on compatibility between viral surface molecules and host receptors.

History of virology developed from observations that some infectious agents could pass through filters that retained bacteria. The field later expanded through plant-virus studies, bacteriophages, electron microscopy and molecular biology.

Exam focus: Draw a labelled generalized virus and explain why viruses are obligate intracellular replicating entities.

2.2 Tobacco Mosaic Virus (TMV)

Definition — TMV: A historically important plant virus that infects tobacco and other susceptible plants and produces characteristic mosaic-like discoloration in leaves.

- Classically studied as a rigid rod-shaped plant virus with a helical arrangement of protein subunits around its RNA.
- Typical symptoms include mottling, chlorosis, distortion and reduced growth depending on host and conditions.
- Importance: TMV became a model system in virology and plant pathology.

Simplified disease progression of TMV

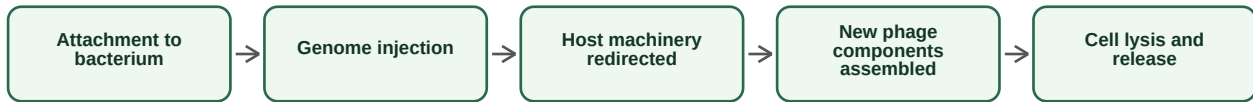


2.3 Bacteriophage

Definition — Bacteriophage: A virus that infects bacteria.

A typical complex phage is described using a head containing nucleic acid, a tail and specialized attachment structures. Different phages vary in morphology.

Simplified lytic cycle



Lytic pathway	Lysogenic pathway
Rapid production of progeny phages	Viral genome can persist in association with host genome/cell lineage
Host cell commonly lyses at the end	Under suitable conditions, a switch to productive replication may occur

Exam focus: Most important phage diagram: labelled head/capsid, genetic material, collar, tail sheath, base plate and tail fibres.

2.4 Mycoplasma

Definition — Mycoplasma: A group of very small bacteria-like prokaryotes characterized by the absence of a rigid cell wall.

- Without a rigid peptidoglycan wall, cell shape is variable (pleomorphism).
- A sterol-containing cell membrane contributes to structural stability in many species.
- Wall-targeting antibiotics are ineffective against organisms lacking the corresponding wall target.
- Some plant diseases are associated with wall-less mollicute groups.

2.5 Archaea

Definition — Archaea: A major domain of prokaryotic organisms distinct from Bacteria, with unique molecular and membrane features.

Feature	Archaea	Bacteria
Membrane lipids	Characteristic ether-linked isoprenoid lipids	Typically ester-linked fatty-acid lipids
Cell wall	No bacterial peptidoglycan; composition varies	Peptidoglycan common in most groups
Habitats	Includes ordinary and extreme environments	Extremely diverse habitats

2.6 Actinomycetes

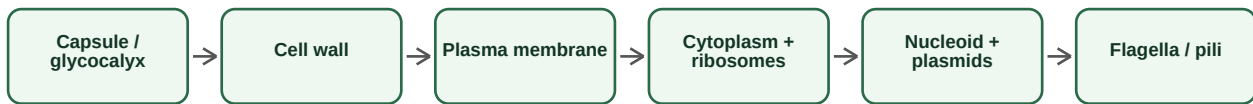
Definition — Actinomycetes: Filamentous, often branching bacteria that may superficially resemble fungal mycelium.

- Important decomposers in soil.
- Many produce secondary metabolites, including medically important antibiotics.
- They demonstrate that morphology alone can mislead classification: filamentous appearance does not necessarily mean a true fungus.

2.7 Bacteria: structure, types, reproduction and economic importance

Definition — Bacterium: A unicellular prokaryotic organism lacking a membrane-bound nucleus and typical membrane-bound organelles.

From outside to inside: generalized bacterial cell



Structure	Main role
Capsule/glycocalyx	Protection, adhesion and surface interaction in many bacteria
Cell wall	Shape and protection against osmotic stress
Plasma membrane	Selective barrier and site of many metabolic processes
Nucleoid	Contains main chromosomal DNA region
Ribosomes	Protein synthesis
Flagella	Motility in many species
Pili/fimbriae	Attachment; some specialized pili participate in DNA transfer

Types based on shape

Type	Shape	Study point
Coccus	Spherical	May occur singly, in pairs, chains or clusters
Bacillus	Rod-shaped	Straight or slightly curved rods
Vibrio	Comma-shaped curved rod	Single curved cells
Spirillum/spirochete	Spiral/helical	Helical forms with different flexibility

Reproduction and genetic variation

Binary fission



Bacteria reproduce mainly by **binary fission**. Genetic variation also arises through mutation and horizontal gene transfer mechanisms such as transformation, transduction and conjugation. These increase variation but are not the same as sexual reproduction in higher organisms.

Economic importance

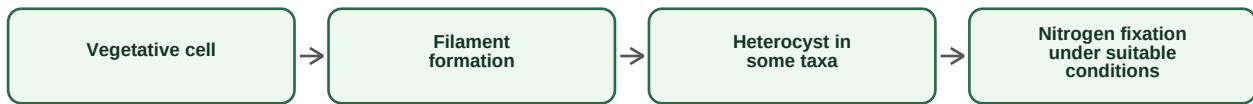
Beneficial roles	Harmful roles
Decomposition, nutrient cycling, food processing, biotechnology, antibiotic production and nitrogen fixation	Disease, food spoilage, crop loss, contamination and some industrial damage

2.8 Cyanobacteria

Definition — Cyanobacteria: Photosynthetic prokaryotes that perform oxygenic photosynthesis and contain chlorophyll a with accessory pigments.

- Historically called blue-green algae, but they are prokaryotic bacteria rather than true algae in modern classification.
- Some filamentous forms differentiate specialized cells such as heterocysts for nitrogen fixation under suitable conditions.
- Ecological roles include primary production and nutrient cycling; excessive growth can cause harmful blooms.

Specialization in selected filamentous cyanobacteria



Exam focus: Unit II priority: generalized virus, TMV, bacteriophage lytic cycle, Mycoplasma vs bacteria, bacterial cell, binary fission and cyanobacteria.